

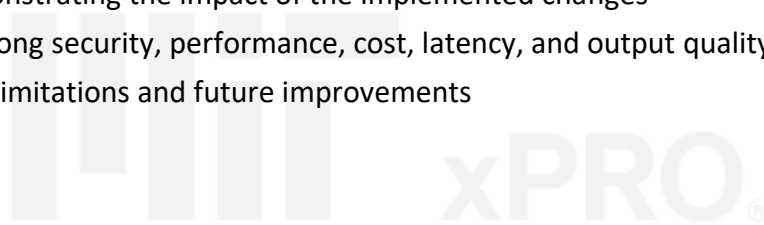
Required Capstone Checkpoint 7.1: Production Readiness – Security and Performance Optimization

Directions

Download this worksheet and save a copy to your device before you begin working. Use this worksheet to document the security safeguards and performance optimizations you implement for your capstone system. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Completion Checklist

- ☐ Implemented one or more safeguards that improve system security and/or reliability
- ☐ Implemented one or more strategies that improve system performance or efficiency
- ☐ Provided evidence demonstrating the impact of the implemented changes
- ☐ Evaluated trade-offs among security, performance, cost, latency, and output quality
- ☐ Reflected on remaining limitations and future improvements



Step 1. System Overview

- Briefly describe your current capstone system.
- Summarize the task your system is designed to complete.

Current system overview

My capstone system is a Wikipedia Retrieval-Augmented Generation (RAG) system that uses both a fixed Context-Aware retrieval workflow and an Agentic Dynamic ReAct workflow. It supports lexical, semantic, hybrid, and Hybrid + Graph retrieval and uses RAGAS-based evaluation to measure answer correctness and failure cases.

Intended task or use case

The system is designed to answer questions using evidence from a collection of Wikipedia articles. It supports:

- Factual retrieval
- Cross-document synthesis
- Exact quotation questions and
- Abstention when the requested information is not available in the corpus.

Step 2. Security and Reliability Improvements

- Identify the safeguard(s) you implemented.
- Explain what risk or failure each safeguard addresses.
- Briefly justify your design decisions.

Implemented safeguard(s)

I implemented a hardened retrieval and answer workflow designed to improve **evidence verification** before retrieved content is used to generate an answer. The safeguard emphasizes distinguishing potentially untrusted corpus content from independently supported evidence and improving abstention when the available evidence is insufficient. I evaluated the hardened workflow against the same five diagnostic questions used for the baseline, covering cross-document synthesis, corpus-poisoning and factual retrieval, out-of-corpus abstention, and quotation fidelity.

Risk(s) addressed

The safeguard addresses the risk of treating retrieved or potentially manipulated corpus content as trustworthy evidence. It also addresses reliability problems involving unsupported claims, cross-document reasoning, and questions that cannot be answered from the available corpus.

Design rationale

I selected this safeguard because Checkpoint 6.1 showed that some configurations did not reliably distinguish potentially untrusted retrieved content from independently supported evidence. The hardening tests were designed to improve evidence handling without replacing the existing RAG architecture. **The results were mixed across configurations**, so the safeguard is treated as a targeted reliability improvement rather than a universal improvement.

Step 3. Performance and Efficiency Improvements

- Describe the optimization strategy or strategies you implemented.
- Explain why you selected these optimizations.

Optimization(s) implemented

I implemented and evaluated two optimization strategies: model selection and retrieval-engine selection. I compared GPT-4o-mini with GPT-5.4-mini and compared the fixed Context-Aware workflow with the Agentic Dynamic workflow using the same five-question diagnostic set. The goal was to identify configurations that provide an appropriate balance of answer quality, token usage, cost, latency, and workflow complexity.

Rationale

I selected these optimizations because the Agentic workflow can provide additional retrieval and reasoning for difficult questions, but it requires substantially more model calls and tokens. The experiments also showed that a more expensive Agentic workflow is not always necessary when Context-Aware produces the same result. This supports selecting the least complex workflow that reliably solves the task.

Step 4. Evidence of Improvement

- Compare system behavior before and after your changes.
- Include representative examples, measurements, screenshots, logs, or outputs.

Before vs. after comparison

Checkpoint 6.1 identified reliability and performance weaknesses that guided the improvements evaluated in Checkpoint 7.1. The earlier evaluation showed that some configurations had difficulty handling potentially untrusted retrieved content and that Agentic processing required substantially more resources than the fixed Context-Aware workflow. In Checkpoint 7.1, the hardened workflow was evaluated against the same five diagnostic questions, while model and retrieval configurations were also compared. With GPT-5.4-mini and Hybrid retrieval, Context-Aware scored 4/5 using 6.3K tokens, \$0.00149 estimated cost, and 8.86 seconds, while Agentic also scored 4/5 but used 32.6K tokens, \$0.00749, and 26.15 seconds. In the cost experiment, Agentic scored 5/5 compared with 4/5 for Context-Aware, but required 40.4K tokens, \$0.00915, and 29.63 seconds versus 9.1K tokens, \$0.00202, and 9.65 seconds. **The hardening results were mixed:** ReAct Hybrid improved from 4/5 to 5/5, while the Lexical test decreased from 5/5 to 4/5 for Agentic and from 4/5 to 3/5 for Context-Aware. Overall, the results show that the **improvements can help with some difficult tasks**, but higher quality can require additional resources and improvements do not consistently transfer across configurations.

Supporting evidence

The Checkpoint 7.1 optimization logs provide evidence from model-selection, workflow-comparison, and hardening experiments. The ReAct Hybrid hardened run passed 5/5 using 31.2K tokens, 32.32 seconds, and 16 workflow steps. The ReAct Hybrid + Graph hardened run passed 4/5 using 31K tokens and 29.47 seconds, with cross-document synthesis remaining a failure. The model-selection experiments showed GPT-5.4-mini achieving 5/5 in one Agentic Hybrid + Graph comparison while using 40.4K tokens and 29.63 seconds, compared with 4/5 for Context-Aware using 9.1K tokens and 9.65 seconds. These results demonstrate that optimization is about selecting an appropriate configuration rather than simply maximizing model or workflow complexity.

Step 5. Trade-off Analysis

- Evaluate the impact of your design decisions.
- Consider security, performance, cost, latency, output quality, usability, or other relevant trade-offs.

Trade-off analysis

The main trade-off is output quality versus cost and latency. Agentic processing can improve difficult questions, but it requires substantially more tokens, model calls, and execution time. A second trade-off is flexibility versus predictability: Context-Aware uses a fixed workflow with lower and more predictable resource usage, while Agentic can perform additional retrieval and reasoning but has greater execution depth. A third trade-off is reliability versus complexity: additional verification can improve some failure cases, but the hardening results were not consistently better across every configuration. These results support using Context-Aware for simpler questions and reserving Agentic processing for questions that require additional reasoning or retrieval.



Step 6. Reflection and Next Steps

- Reflect on the effectiveness of your improvements.
- Identify remaining limitations and describe what you would improve next.

Effectiveness of improvements

The improvements were partially effective. Checkpoint 6.1 identified reliability weaknesses in evidence handling and showed that Agentic processing could improve difficult tasks but required substantially more tokens and model steps than the Context-Aware workflow. These findings guided the Checkpoint 7.1 hardening and optimization efforts. In 7.1, GPT-5.4-mini produced strong Agentic results with lower token usage and latency than GPT-4o-mini in several experiments, while Context-Aware remained substantially more efficient for simpler tasks. The hardening results were mixed: ReAct Hybrid improved from 4/5 to 5/5, but the October 6 Lexical test decreased from 5/5 to 4/5 for Agentic and from 4/5 to 3/5 for Context-Aware. Overall, the improvements addressed some of the weaknesses identified in 6.1, but the results show that no single configuration consistently maximized reliability, quality, cost, and latency.

Remaining limitations

Several limitations remain. Cross-document synthesis continues to be a recurring weakness, particularly in some Semantic and Hybrid + Graph configurations, which was also identified in earlier evaluations. The 7.1 hardening results show that improving one configuration does not necessarily improve all retrieval strategies and may sometimes reduce performance. The diagnostic test set contains only five questions, so it is useful for identifying specific failure modes but is not large enough to establish broad production reliability. In addition, query-based routing has not yet been validated on a larger and more representative workload, so the system still needs further testing to determine when Context-Aware or Agentic processing should be selected.

Future improvements

The next improvement would be to implement and evaluate query-based routing based on the complexity of each question. Simple factual questions could use the Context-Aware workflow because it provides lower cost, lower latency, and more predictable execution, while questions requiring cross-document synthesis, additional retrieval, or more complex reasoning could be routed to the Agentic workflow. I would also like to expand the evaluation set beyond the current five diagnostic questions, with additional cross-document, quotation, corpus-poisoning, and out-of-corpus tests. Finally, I would also like to continuously monitor answer quality, cost, latency, and failure categories to determine whether the routing strategy improves overall system performance and reliability before production deployment.



Supporting Materials

Use this section to list or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
Test cases	Test_Questions_Manually_Failure.json — five Checkpoint 7.1 diagnostic questions covering cross-document synthesis, corpus-poisoning/factual retrieval, out-of-corpus abstention, and quotation fidelity. https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE
Performance measurements	checkpoint_7_1_optimization_performance.log — pass rate, tokens, estimated cost, latency, model calls, and workflow steps. https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE
Example queries and outputs	Diagnostic examples include the Einstein corpus-poisoning question, the Mars population out-of-corpus question, the Einstein Nobel claim verification question, and the exact Nobel Prize quotation question. https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE
Screenshots or logs	GitHub logs: https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE
Code/notebook/repository	https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE

Supporting material	Location, link, or note
Other supporting evidence	Checkpoint 5.1 and 6.1 evaluation results provide the baseline for the Agentic-vs-fixed workflow trade-off and the security/reliability issues addressed in Checkpoint 7.1. https://github.com/miguelangelherrera77-blip/MIT_CAPSTONE

End of Required Capstone Checkpoint 7.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.

